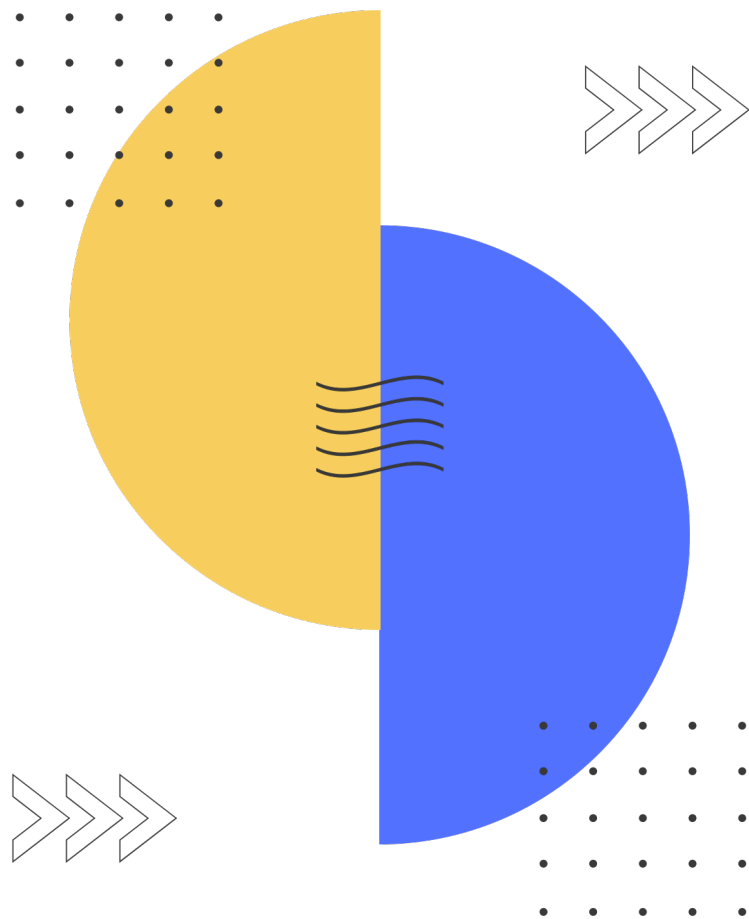




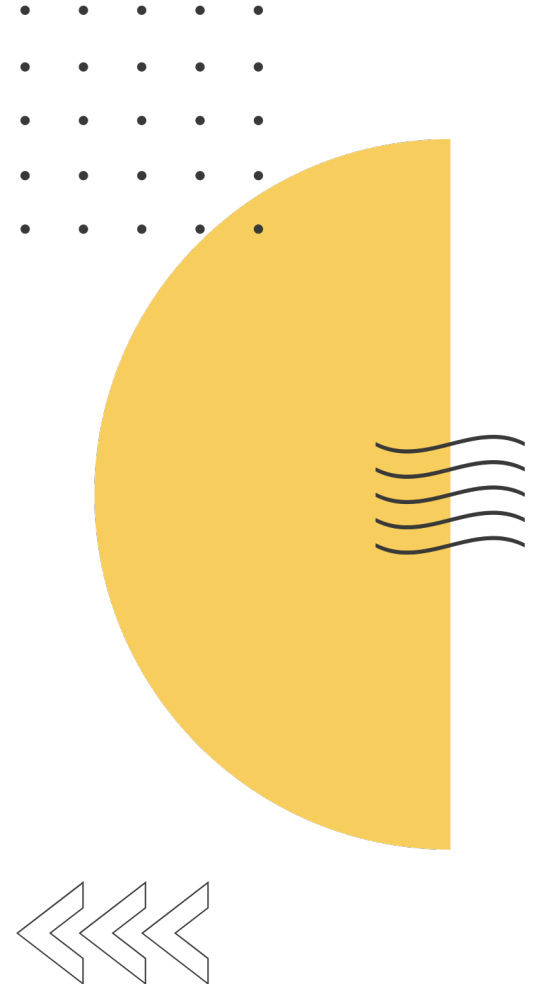
AI ACT

Prof. Francesca Lagioia
Prof. Giovanni Sartor
Dr. Rūta Liepiņa
ALMA-AI, University of Bologna
May 2026

OUTLINE

PART I: AI ACT

- Context and background
- Definitions and scope
- Risk-based regulation approaches
- Risk levels
- Implementation



CONTEXT AND BACKGROUND



Dangers of unregulated artificial intelligence

WIRED SECURITY POLITICS THE BIG STORY BUSINESS SCIENCE CULTURE IDEAS MERCH REVIEWS

KHARI JOHNSON BUSINESS DEC 2, 2021 7:00 AM

The Movement to Hold AI Accountable Gains More Steam

A New York City law requires algorithms used in hiring to be “audited” for bias. It’s the first in the US—and part of a larger push toward regulation.



The New York Times

Artificial Intelligence > Microsoft Moderates A.I. Spending A.I. Forecast A.I.’s Super Bowl A.I. Hallucinations Quiz

Five Ways A.I. Could Be Regulated

There is little agreement among regulators and lawmakers around the world on how artificial intelligence should — or even could — be controlled.

Share full article



INDEPENDENT
HIGH-LEVEL EXPERT GROUP ON
ARTIFICIAL INTELLIGENCE
SET UP BY THE EUROPEAN COMMISSION

ETHICS GUIDELINES
FOR TRUSTWORTHY AI

2019

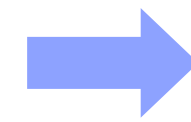
CONTEXT AND BACKGROUND

To counter AI risks, while preserving scientific research and the beneficial uses of AI, a number of initiatives have been undertaken in order to design an ethical and legal framework for “human-centred AI.

2016 **initial reports** on how to prepare for the future of AI were released:

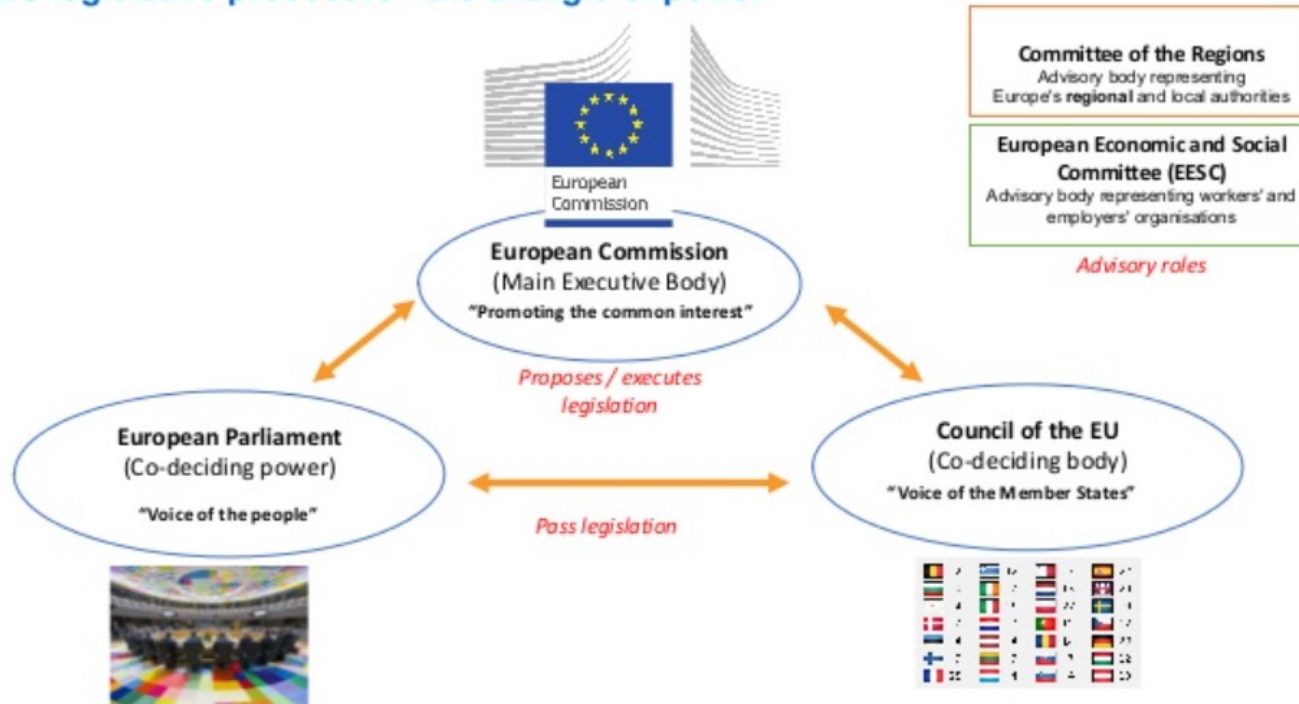
- The White House Office of Science and Technology Policy (OSTP)
- The European Parliament’s Committee on Legal Affairs
- In UK, the House of Commons’ Science and Technology Committee
- Multiple expert committees have subsequently produced reports and policy documents, including:
 - ❑ High-Level Expert Group on Artificial Intelligence appointed by the European Commission
 - ❑ The expert group on AI in Society of the Organisation for Economic Co-operation and Development (OECD),
 - ❑ The select committee on Artificial Intelligence of the United Kingdom (UK) House of Lords,
 - ❑ The Council of Europe (CEPEJ)

LAW MAKING IN THE EU



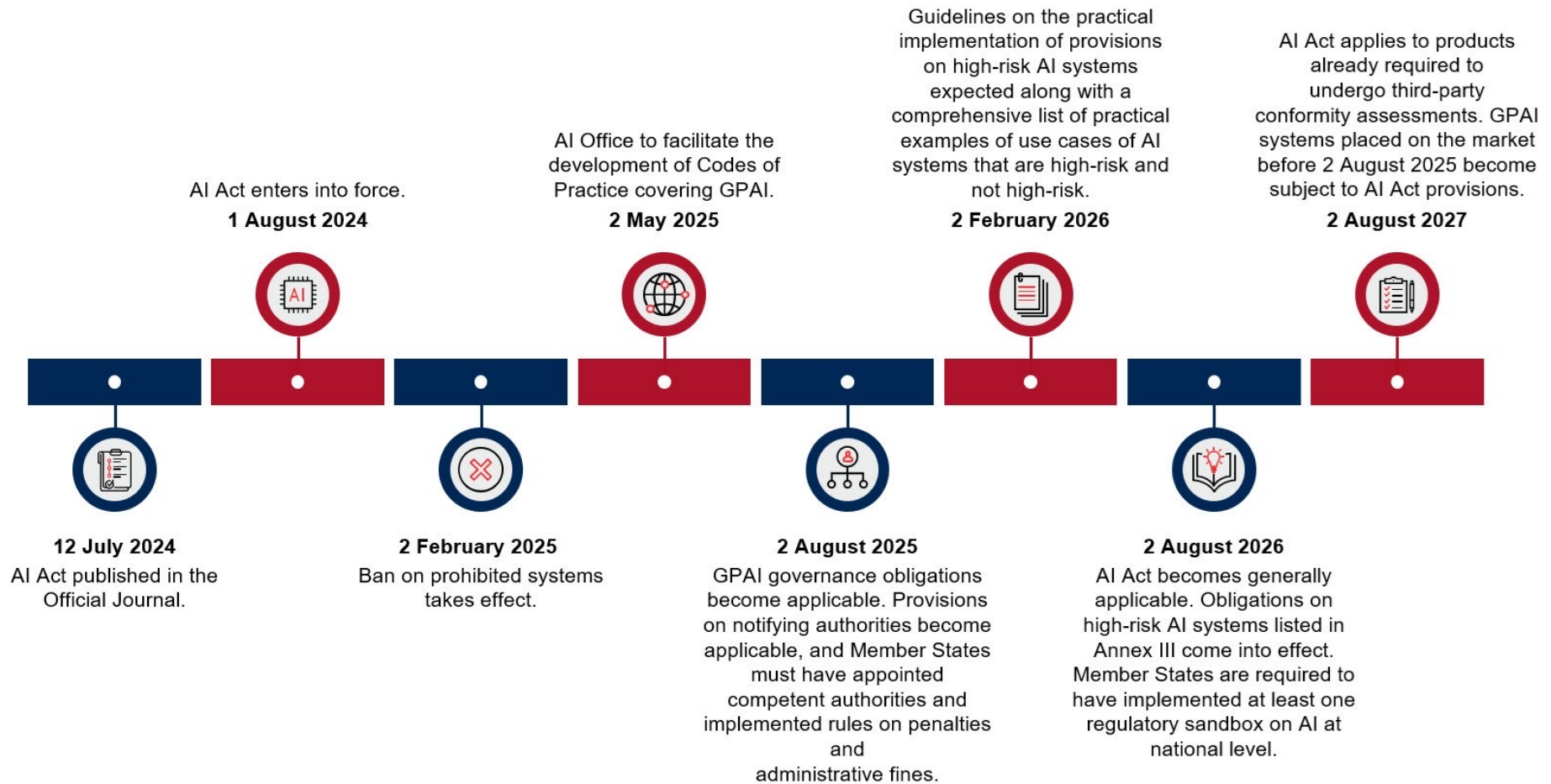
AI ACT
proposal (2021
-> Approval
2024)

EU legislative procedure - the triangle of power



EU legislative process: <https://www.atsep.eu/global/>

Timeline of provisions

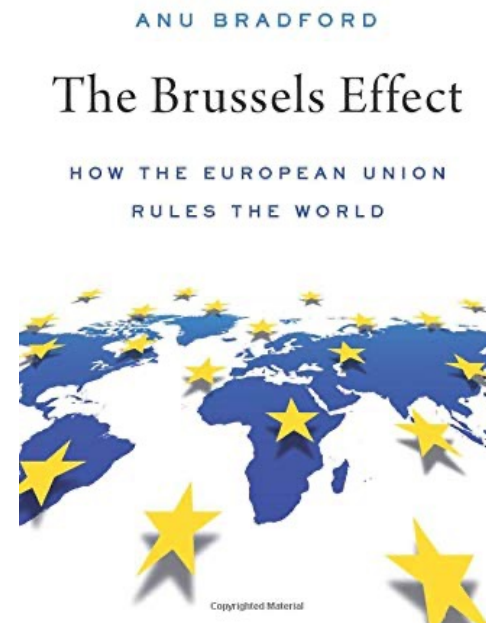


THE GENESIS OF AI ACT: Innovation and Fundamental Rights

1) Ethics guidelines for trustworthy AI (2019)

2) EU white paper on AI, 2020

- research and deployment of AI should be **promoted**, so that the EU is **competitive** with the US and China.
- the deployment of AI technologies should be **consistent with the EU fundamental rights and social values**.



GENERAL OVERVIEW

- An AI system is a product that has (incorporates) a purpose.
- Certain purposes are assumed to entail certain specific 'risks'.
- To AI systems with different purposes and thus different levels of risk, correspond different levels of more or less stringent rules, aimed in particular at those who develop and/or implement the system

GOALS AND TWOFOL APPROACH

Product safety approach

- Risk-based approach (top-down)
- Technical requirements
- Conformity assessment
- Market surveillance/Public Enforcement



Fundamental rights

- Risk-based approach (bottom-up)
- Fundamental rights impact assessment (*in some cases*)
- No new individual rights



General Principles of trustworthy AI (Recital 27)

The 7 principles of **trustworthy AI (developed by the High Level expert group)** should be translated, when possible, in the design and use of AI systems.

All operators falling under this Regulation shall make their best efforts to develop and use AI systems and general purpose models in accordance with the general principles of:

- human agency and oversight
- technical robustness and safety (data, design, methods, etc.)
- privacy and data governance
- transparency
- diversity, non-discrimination and fairness
- social and environmental well-being
- accountability



Not binding but useful for the interpretation of the act and voluntary codes of conduct

AI ACT: OVERVIEW

 EU Artificial Intelligence Act

The Act

Implementation

Context

About us

Newsletter

 EN

Table of Contents

The EU AI Act consists of 12 main titles. Each title contains a set of Articles.

Chapter I: General Provisions

Chapter II: Prohibited AI Practices

Chapter III: High-Risk AI System

Chapter IV: Transparency Obligations for Providers and Deployers of Certain AI Systems

Chapter V: General-Purpose AI Models

Chapter VI: Measures in Support of Innovation

Chapter VII: Governance

Chapter VIII: EU Database for High-Risk AI Systems

Chapter IX: Post-Market Monitoring, Information Sharing and Market Surveillance

Chapter X: Codes of Conduct and Guidelines

Chapter XI: Delegation of Power and Committee Procedure

Chapter XII: Penalties

Chapter XIII: Final Provisions

Annexes

Annexes provide supplementary information alongside the Regulation.

Annex I: List of Union Harmonisation Legislation

Annex II: List of Criminal Offences Referred to in Article 5(1), First Subparagraph, Point (h)(iii)

Annex III: High-Risk AI Systems Referred to in Article 6(2)

Annex IV: Technical Documentation Referred to in Article 11(1)

Annex V: EU Declaration of Conformity

Annex VI: Conformity Assessment Procedure Based on Internal Control

Annex VII: Conformity Based on Assessment of the Quality Management System and an Assessment of the Technical Documentation

Annex VIII: Information to be Submitted upon the Registration of High-Risk AI Systems in Accordance with Article 49

Annex IX: Information to be Submitted upon the Registration of High-Risk AI Systems Listed in Annex III in Relation to Testing in Real World Conditions in Accordance with Article 60

Annex X: Union Legislative Acts on Large-Scale IT Systems in the Area of Freedom, Security and Justice

Annex XI: Technical Documentation Referred to in Article 53(1), Point (a) - Technical Documentation for Providers of General-Purpose AI Models

Annex XII: Transparency Information Referred to in Article 53(1), Point (b) - Technical Documentation for Providers of General-Purpose AI Models to Downstream Providers that Integrate the Model into Their AI System

Annex XIII: Criteria for the Designation of General-Purpose AI models with Systemic Risk Referred to in Article 51

Recitals

Recitals provide context about how an article should be interpreted or implemented.

1	2	3	4	5	6	7	8	9
10	11	12	13	14	15	16	17	18
19	20	21	22	23	24	25	26	27
28	29	30	31	32	33	34	35	36
37	38	39	40	41	42	43	44	45
46	47	48	49	50	51	52	53	54
55	56	57	58	59	60	61	62	63
64	65	66	67	68	69	70	71	72
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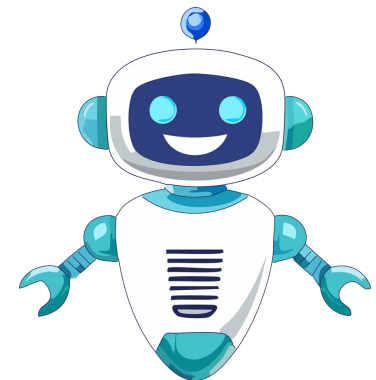
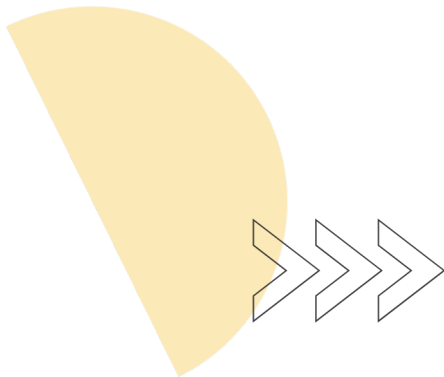
<https://artificialintelligenceact.eu/ai-act-explorer/>

Object: AI SYSTEM Definition

'AI system' means a **machine-based system** that is designed to operate with **varying levels of autonomy** and that may exhibit **adaptiveness** after deployment, and that, for explicit or implicit objectives, **infers**, from the input it receives, how to generate **outputs** such as predictions, content, recommendations, or decisions that can influence physical or virtual environments [Art. 3 (1)]



+ **Recital 12:** [...] based on key characteristics of AI systems that **distinguish it from simpler traditional software systems** or programming approaches and should not cover systems that are based on the rules defined solely by natural persons to automatically execute operations (e.g., ability to infer).

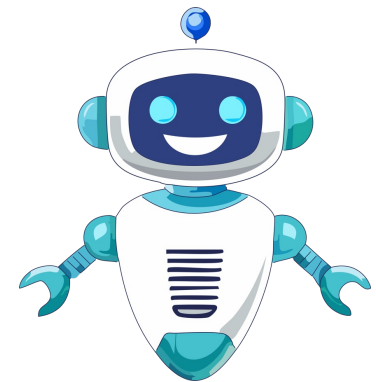
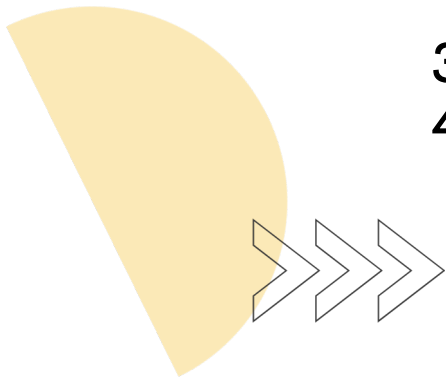


Critics: AI SYSTEM Definition

The definition of an “AI system” in Article 3(1) of the AI Act, clarified in Recital 12, has been modified over time, due to several criticisms from academics, legal scholars, computer scientists, and critical technology studies experts. These critiques revolve around 4 main axes:



1. Overly Broad and Generic Definition
2. Ambiguity Between Symbolic and Sub-Symbolic AI
3. Focus on Function Rather than Technology
4. Risk of “Performative Regulation”

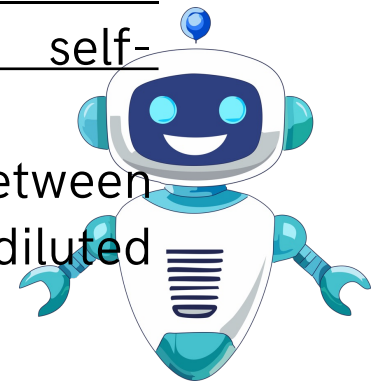


Critics: AI SYSTEM Definition

1. Overly Broad and Generic Definition: it is too expansive, potentially encompassing nearly every form of advanced software. (Veale, M., & Borgesius, F. Z. (2021); AlgorithmWatch (2021).

- It lacks precise clarification of terms such as "**machine**", "**inference**", "**adaptiveness**", or "**autonomy**".
- It includes any system capable of generating outputs that **influence virtual or physical environments** — which could cover even basic recommendation systems or rule-based engines with minor self-configuration.

 **Key concern:** this leads to **overinclusiveness**, blurring the line between classical software, advanced statistics, and "genuine AI". The result is a diluted regulatory scope and an ineffective governance strategy.

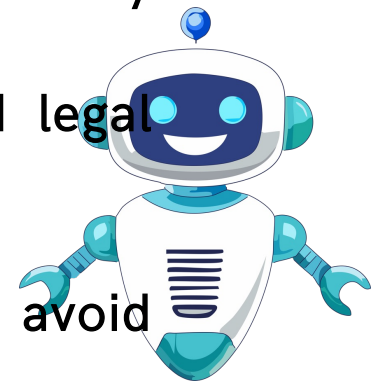


Critics: AI SYSTEM Definition

2. Ambiguity Between Symbolic and Sub-Symbolic AI: it insufficiently distinguishes between rule-based (symbolic) and learning-based (sub-symbolic) systems, leaving hybrid or borderline technologies in a legal grey zone, although Recital 12 attempts to exclude software that follows rules solely defined by humans. (Hacker, P. (2022); Wachter, S. & Mittelstadt, B. (2022))

- In practice, however, many hybrid systems (rule-based + ML) defy simple classification.
- The blurred boundary could lead to regulatory grey areas and legal uncertainty.

 Risk: developers might self-classify their systems as “not AI” to avoid regulatory obligations, exploiting this ambiguity.



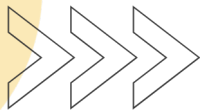
Critics: AI SYSTEM Definition

3. Focus on Function Rather than Technology: the definition focuses on **system outputs** (e.g., predictions, decisions, content) **rather than the technical nature of the system itself**. (Edwards, L. (2022); AI Now Institute (2022)).



➤ This teleological approach is flexible but fragile: it relies on **subjective impact assessments rather than objective technological criteria**.

 Some researchers advocate for **more technically grounded definitions**, capable of distinguishing between deterministic logic, machine learning, and deep neural architectures.



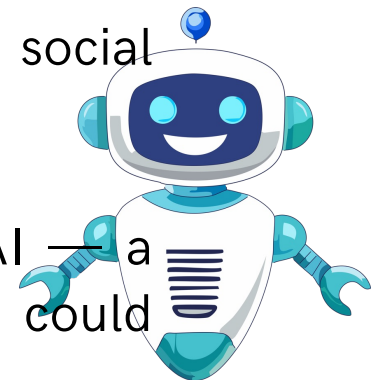
Critics: AI SYSTEM Definition

4. Risk of "Performative Regulation" : scholars from critical disciplines (e.g., STS and AI ethics) argue that this definition doesn't describe AI as it is, but rather AI as institutions want to regulate it (Jaton, F. (2021); Binns, R. (2020); Yeung, K. (2018))

- It is a **legal construct** designed to justify regulatory intervention, which **may fail to reflect the actual industrial or technical use** of the term.

This may create an epistemic disconnect between legal, technical, and social understandings of AI.

📢 In other words, the EU is constructing a particular kind of legal AI — a regulatory reality that may not match the technological one — which could result in failure to capture opaque or emerging practices.



Object : GENERAL PURPOSE AI

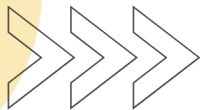
'general-purpose AI model' means an AI model,

- Including where such an AI model is trained with a large amount of data using self-supervision at scale, that
- that displays significant generality and is capable of competently performing a wide range of distinct tasks [...]and
- that can be integrated into a variety of downstream systems or applications [...]; [Art 3 (63)]



'general-purpose AI system' means an AI system

- which is based on a general-purpose AI model and
- which has the capability to serve a variety of purposes, both for direct use as well as for integration in other AI systems;



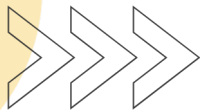
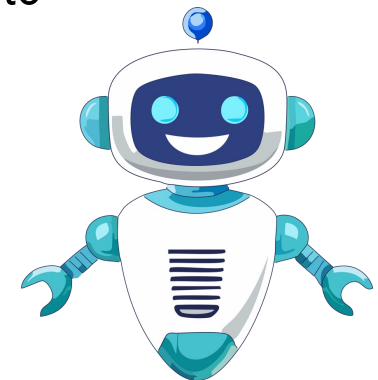
Object : GENERAL PURPOSE AI

Balancing Innovation and Accountability

The definition of general-purpose AI models (GPAI models) was part of a political compromise: rather than banning or heavily regulating generative AI outright, EU legislators chose to distinguish between:

- **GPAI models** → abstract, upstream components, subject to **obligations such as transparency, technical documentation, and risk management.**
- **GPAI systems** → concrete applications built on top of those models, **potentially subject to classification as high-risk under the AI Act.**

This layered approach allows responsibility to be distributed across the AI value chain: from the model developer, to the system provider, and down to the end user.

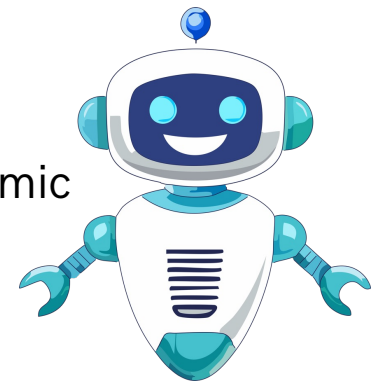
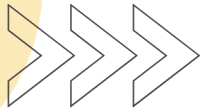


Object: SYSTEMIC RISK

‘**systemic risk**’ means a risk that is specific to the **high-impact capabilities of general-purpose AI models**, having a significant impact on the Union market

- due to their reach, or
- due to actual or reasonably foreseeable **negative effects on public health, safety, public security, fundamental rights, or the society as a whole, that can be propagated at scale across the value chain;**

- Article 55: Obligations for Providers of General-Purpose AI Models with Systemic Risk



Object: SYSTEMIC RISK

Systemic risk refers to a type of **risk that is not confined to a single actor, event, or domain**, but has the **potential to spread and destabilize an entire system — whether economic, social, technological, or ecological**. It is an interconnected, non-linear, and high-impact risk, often hard to detect or contain once triggered.



Key Characteristics of Systemic Risk:

- **Domino effect:** a local or sector-specific issue can cascade through the entire system.
- **Interdependence:** the components of the system are so interconnected that a failure in one part can propagate.
- **Opacity and unpredictability:** systemic risks tend to be emergent, complex, and poorly captured by conventional risk assessment tools.

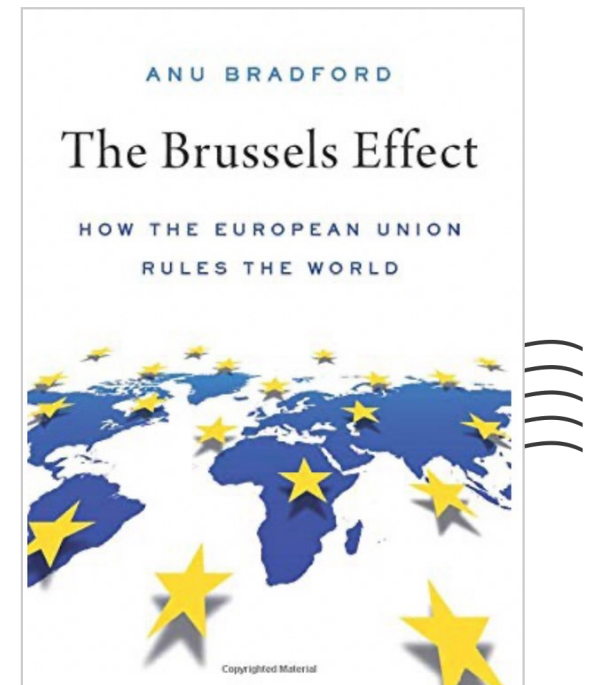


SCOPE I

The Regulation applies to

- a) **Providers** placing on the market or putting into service AI systems or general-purpose AI models in the Union, irrespective of whether those providers are established within the Union or in a third country;
- b) **Deployers** (e.g., public or private entities, natural persons for professional activities) of AI systems located within the Union;
- c) **Providers and deployers** of AI systems that are located in a third country, **where the output produced by the system is used in the Union** (e.g., an AI-based medical diagnosis app developed in the USA, whose output is used by a European hospital)

[Art. 2]

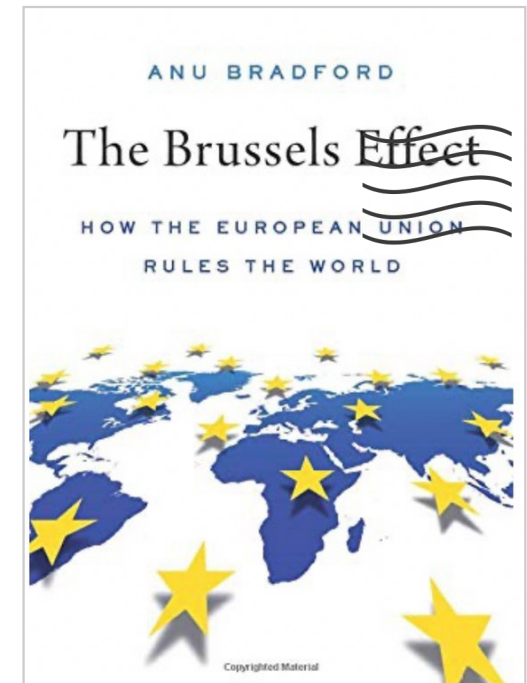


SCOPE II

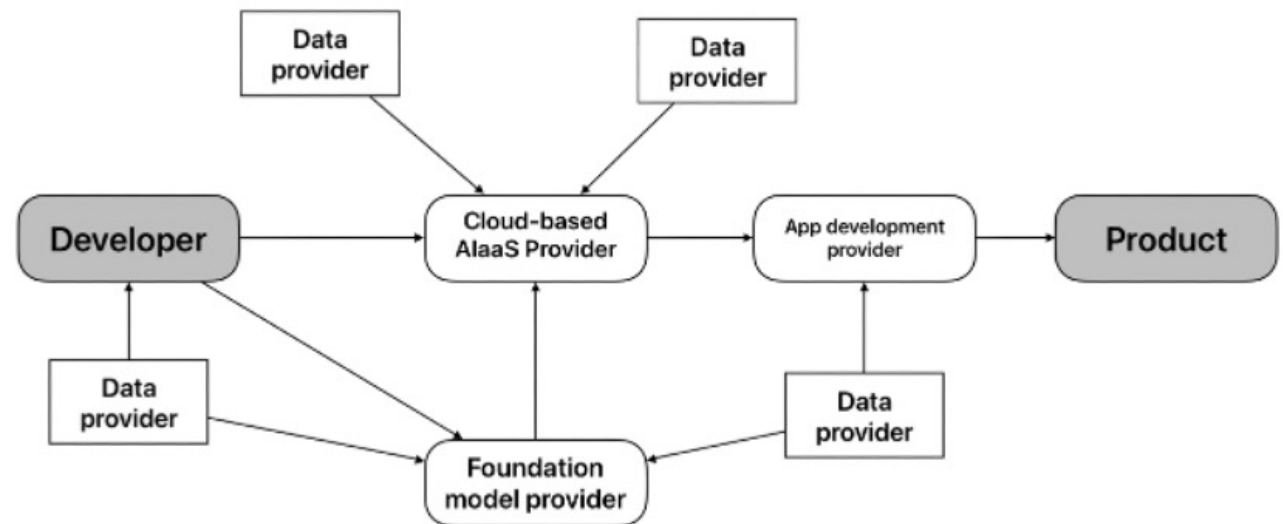
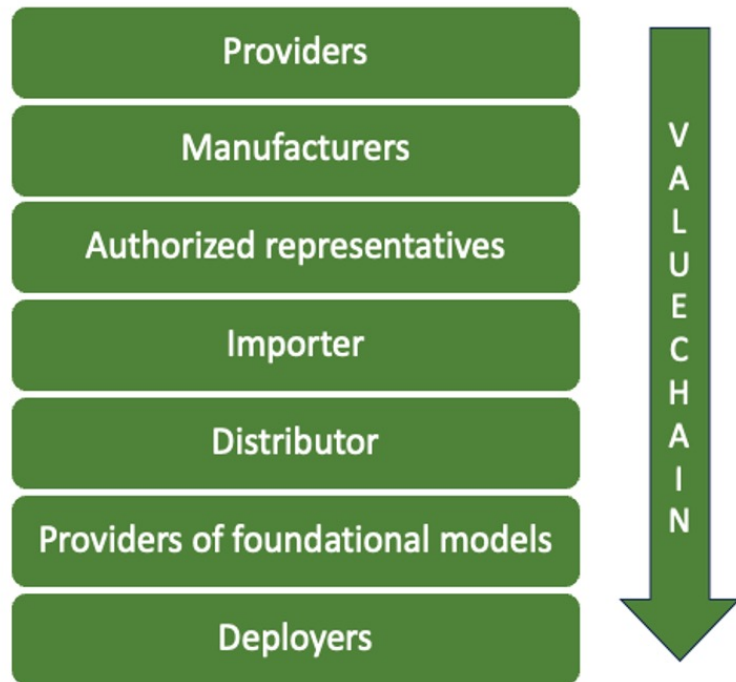
The Regulation applies to

- d) **importers and distributors** of AI systems;
- (e) **product manufacturers** placing in the market or putting into service an AI system together with their product and under their own name or trademark; (e.g., a car manufacturer that integrates an autonomous driving system into a vehicle marketed in the EU).
- (f) **authorised representatives** of providers, which are not established in the Union; (often appointed to represent the provider's legal interests within the EU.)
- (g) **affected persons** that are located in the Union (Individuals whose rights or interests are affected by the use of an AI system. Even without being users or developers)

[Art. 2]



ACTORS IN THE VALUE CHAIN



EXCLUDED AREAS

- Military, defence, and national security uses are fully excluded
- Scientific research and development activities are not covered
- Pre-market development and testing is excluded, except for real-world testing
- International law enforcement cooperation may be excluded IF fundamental rights safeguards are in place



Inspiration: Safety + Fundamental Rights/Values

Product safety approach

- Risk-based approach
- Focus on consumers' health and safety
- Technical requirements
- Conformity assessment
- Market surveillance/Public Enforcement



Fundamental rights/Values

- Assessment includes impact on fundamental rights and values
- Yet, no new individual rights



AIA as a risk regulation

- **Aim:** prevent harm proactively, by minimizing risks
- **Risks:** adverse impacts on health, safety, fundamental rights, and values (environment, rule of law, democracy)
- **Stringency of regulations:** should be proportionate to level of risk
- **Risk assessment:** evidence based evaluation, according to data, context, and intended use
- **Interventions:** risks should be minimized through mitigation measures
- **Residual risks should be acceptable:** proportionate balance of risks vs expected benefits
- **Accountability:** developers/deployers should address risks and document their efforts
- Development and deployment of useful technologies should not be hindered



Will AIA be successful as a risk regulation?

WHAT IS RISK?

- Risk1(hazard) = potential **harmful event**
 - E.g.: The event of an AV is hitting pedestrians in urban areas (Here, risk corresponds to the type of threat (e.g., physical harm, discrimination, unlawful surveillance), **regardless of its probability or impact**)
- Risk2 = The **probability of a harmful event** (Art. 4 (2)) => quantitative/statistical approach
 - E.g.: There is a 1/1,000000 probability that an AV will hit a pedestrian over one year of use
- Risk3 = **Combination of probability and severity**.
 - E.g. The risk of an accident by an AV is medium level since its harm is highly improbable but very severe
- What if the probability of harm cannot be determined: uncertainty vs known probability (known unknowns, e.g., violations of fundamental rights by AI)?
- What if it is not even possible to exhaustively identify possible harmful events (unknown unknowns, e.g., LLMs effects on the future of creativity)?

RISK LEVELS



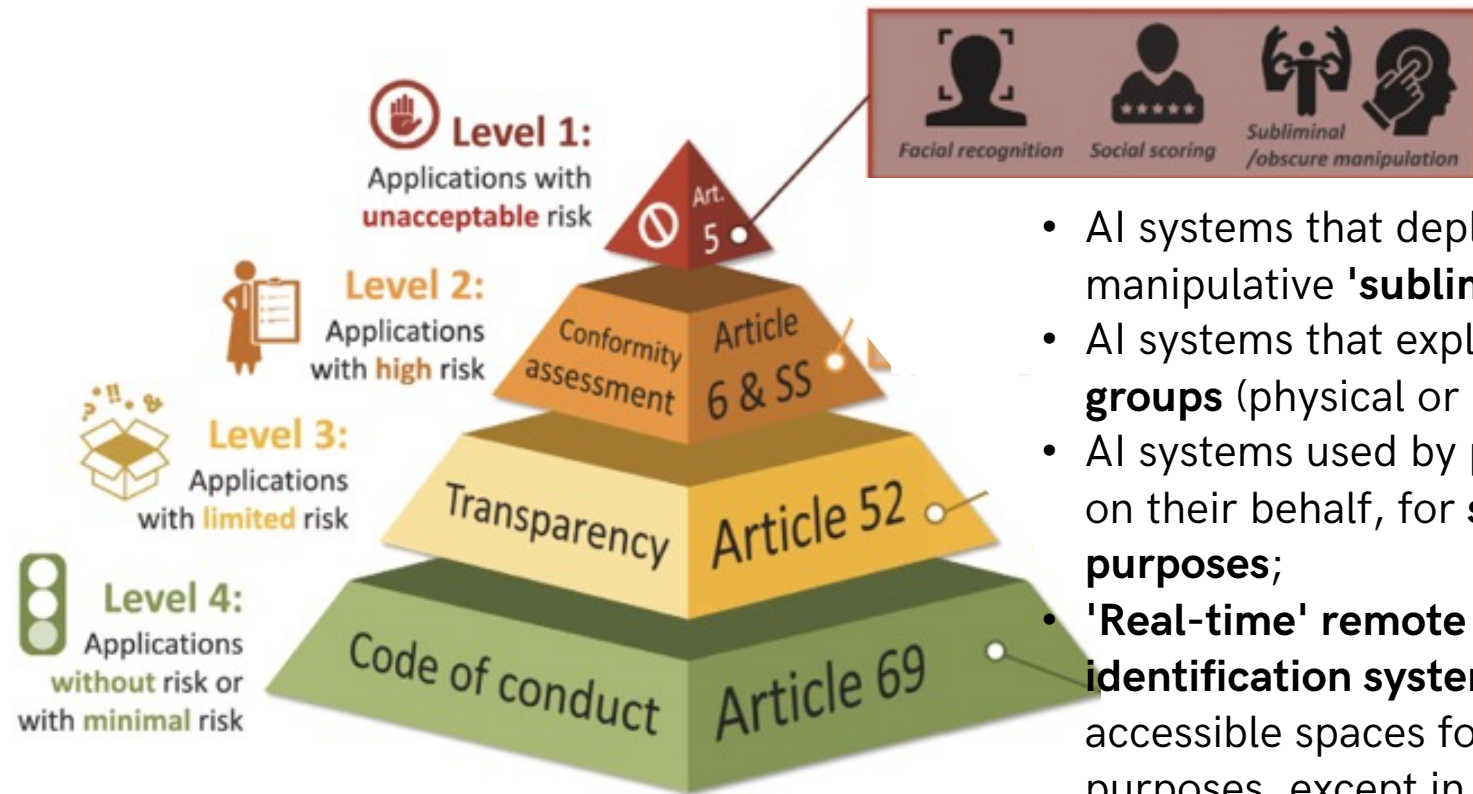
The AIA distinguishes two main classes of AI applications.

- Certain kinds of applications are classified as “**unacceptable risk**”: they are prohibited.
- Certain kinds of applications are classified as **high-risk**: they are regulated and submitted to certification

Moreover the AIA:

- introduces **transparency obligations for all systems** which may be mistaken for humans, or whose outputs may be mistaken for human generated
- supports the creation of **voluntary codes of conduct**, for all systems

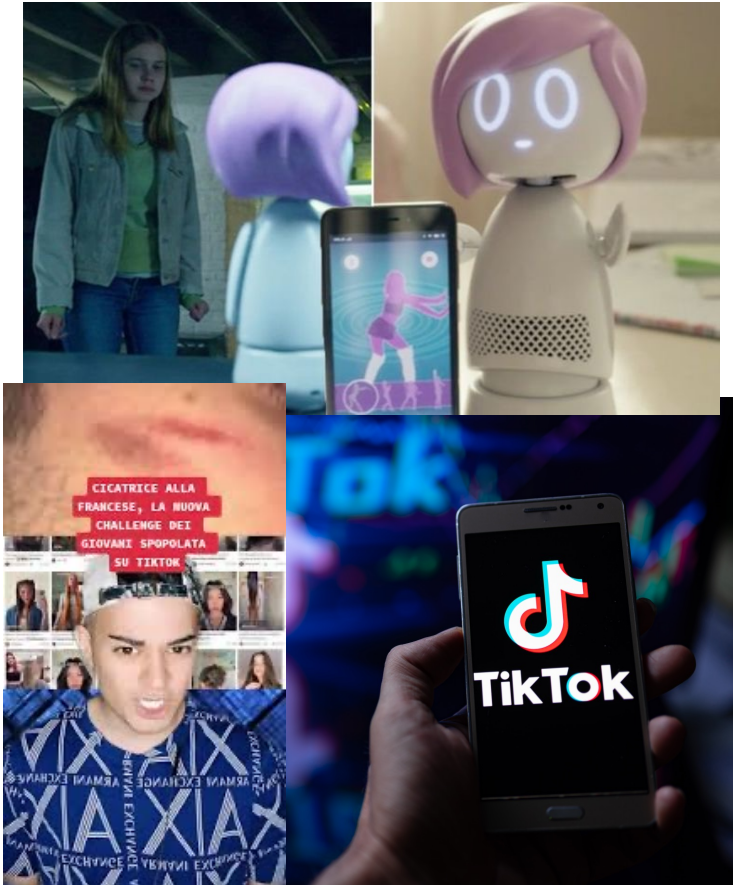
RISK LEVELS: unacceptable risk



- AI systems that deploy harmful manipulative '**subliminal techniques**';
- AI systems that exploit specific **vulnerable groups** (physical or mental disability);
- AI systems used by public authorities, or on their behalf, for **social scoring purposes**;
- '**Real-time**' remote biometric **identification systems** in publicly accessible spaces for law enforcement purposes, except in a limited number of cases.

Prohibited AI Practices

- Systems that **manipulate behavior**, especially targeting specific groups or vulnerable populations.
- Systems **inferring emotions** in workplaces or educational institutions, except for medical or safety reasons.
- Systems that implement **social scoring**, which involves evaluating individuals based on their behavior, socio-economic background, or personal traits.
- Systems **assessing the risk of an individual committing criminal offenses** solely based on profiling or personality traits.
- **Biometric categorisation systems inferring sensitive attributes** (race, political opinions, trade union membership, religious or philosophical beliefs, sex life, or sexual orientation).
- Systems capable of **'real-time' remote biometric identification (RBI)** in publicly accessible spaces for law enforcement (but many exceptions).



RISK LEVELS: high risk



Annex II legislation covers:

- Machinery
- Toys
- Lifts
- Radio Equipment
- Personal watercraft
- Medical devices
- Equipment used in potentially explosive atmosphere
- Pressure equipment
- Personally protective equipment
- Cableway installations
- Appliances burning gaseous fuels
- Civil aviation security
- 2- or 3-wheel vehicles and quadricycles
- Agricultural and forestry vehicles
- Marine equipment
- Interoperability of the rail system
- Motor vehicles and their trailers
- Civil aviation

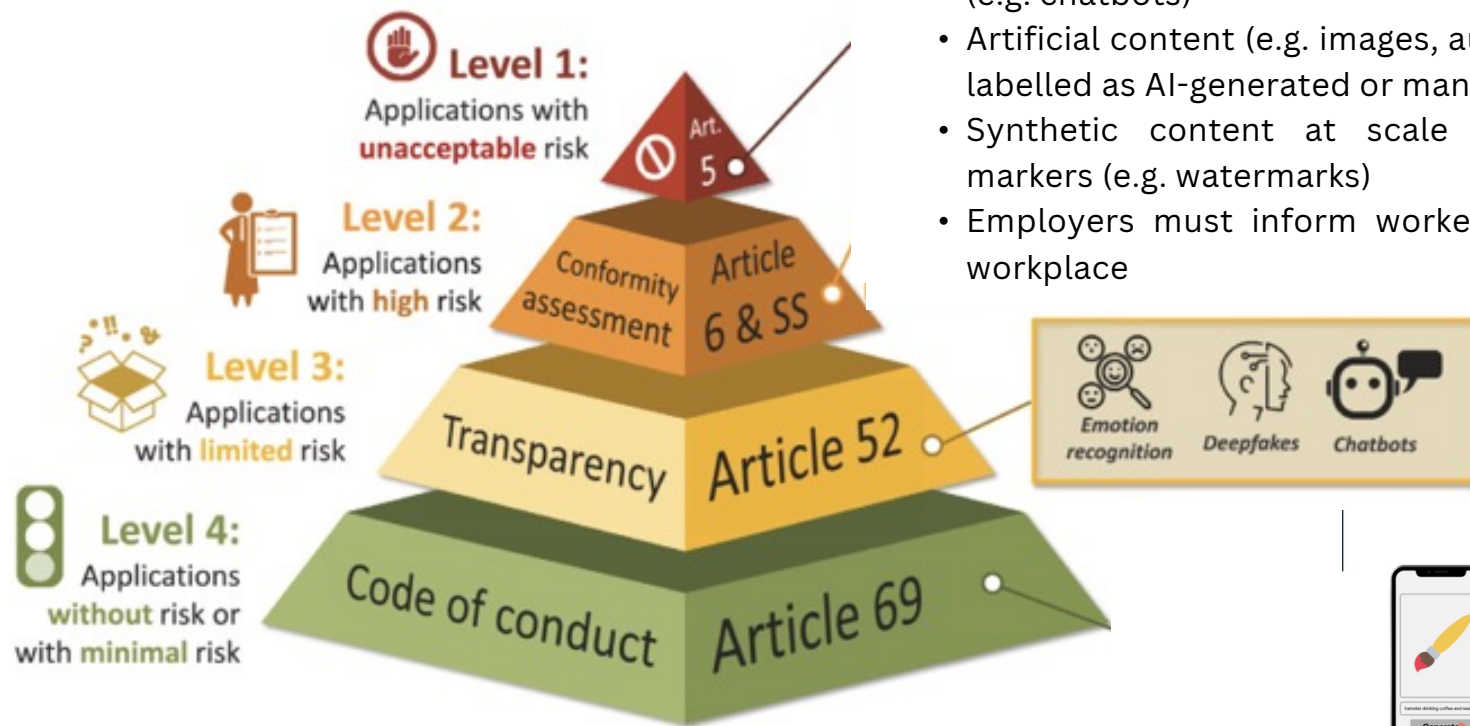
Annex III categories of purposes:

- Biometric identification and categorisation of natural persons
- Management and operations of critical infrastructure
- Education and vocational training
- Employment, workers management and access to self-employment
- Access to and enjoyment of essential private services and public services and benefits
- Law enforcement
- Migration, asylum and border control management
- Administration of justice and democratic processes

What “new” AI systems classified as high risks (Annex III)?

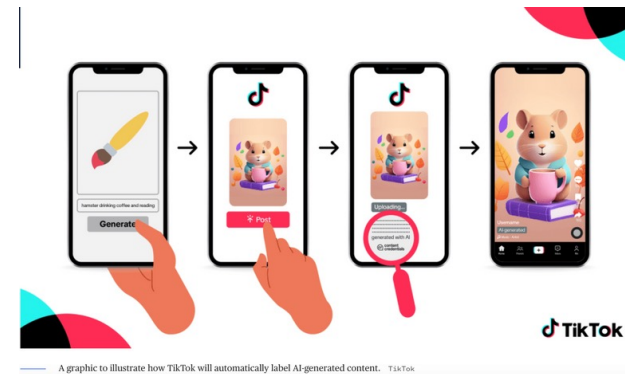
- AI used for **remote biometric identification, emotion recognition and biometric categorisation** (e.g. AI system to retroactively identify a shoplifter)
- AI **safety components in critical infrastructures** (e.g. transport), the failure of which could put the life and health at risk
- AI used in **education institutions**, that may determine the access to education and course of someone’s professional life (e.g. scoring of exams)
- **AI-based safety components** of products (e.g. AI application in robot-assisted surgery)
- AI tools for **employment, management of workers and access to self-employment** (e.g. CV-sorting software for recruitment)
- Certain AI use-cases utilised to give **access to essential private and public services** (e.g. credit scoring denying citizens opportunity to obtain a loan)
- AI use-cases in **law enforcement** that may interfere with fundamental rights (e.g. evaluation of the reliability of evidence)
- AI use-cases in **migration, asylum and border control management** (e.g. automated examination of visa applications)
- AI solutions used in the **administration of justice and democratic processes** (e.g. AI solutions to prepare court rulings)

RISK LEVELS: limited risk/ transparency



- Applies to interactive AI systems like chatbots, emotion recognition, biometric categorisation, and deepfakes
- Users must be informed when interacting with an AI system (e.g. chatbots)
- Artificial content (e.g. images, audio, video) must be clearly labelled as AI-generated or manipulated
- Synthetic content at scale must include detectable markers (e.g. watermarks)
- Employers must inform workers when AI is used in the workplace

For example,
TikTok automated labelling:



NB: Level 3 is not really a distinct category of systems: Art. 52 applies any system, high risk or not, which can be mistaken for a human or whose outcomes may be mistaken for human generated

RISK LEVELS: no risk/ minimal risk

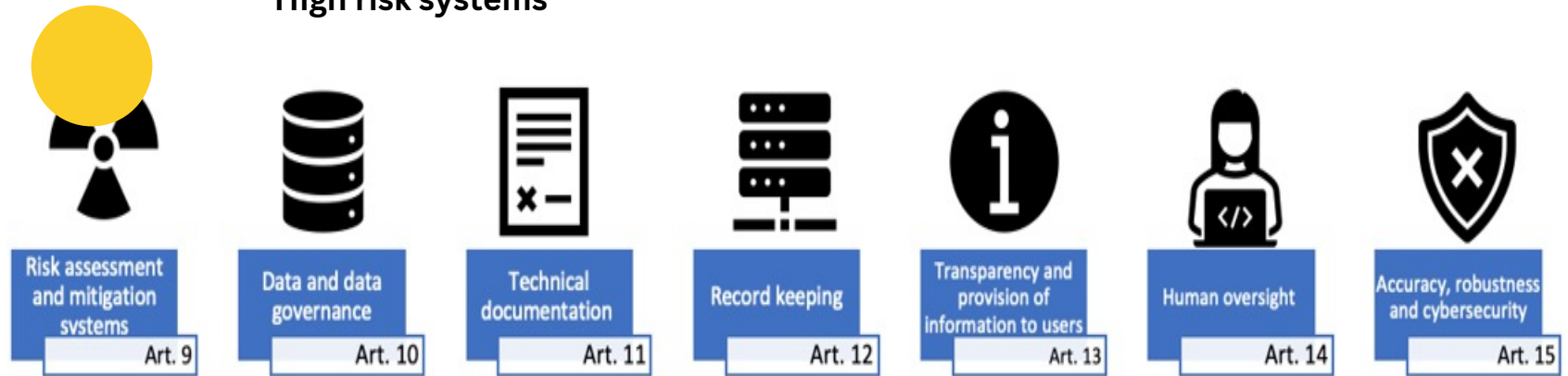


- Minimal risks.
- Systems presenting minimal risk for people (e.g. spam filters) will not be subject to further obligations beyond currently applicable legislation (e.g., GDPR).

NB: not a real category: just whatever is not classified as unacceptable-risk, or high-risk and is not susceptible to mislead humans about its artificial nature.

MANDATORY REQUIREMENTS

High risk systems



General Purpose AI (ex. foundational models)



Foundation models

Performance, predictability, interpretability

Reduce energy impact

Registration in EU database



Generative AI

Transparency requirements

Ensuring lawful content

Whether copyrighted are used

MANDATORY REQUIREMENTS–High Risk

- **Art 9: Risk management system:** adequate **risk assessment and mitigation** systems;
- **Art 10: data and data governance:** High quality of the datasets feeding the system to **minimise errors and discriminatory outcomes**;
- **Art 11: Technical Documentation:** Detailed documentation providing all information necessary **on the system and its purpose for authorities to assess its compliance**;
- **Art 12: Record –keeping: Logging of activity to ensure traceability of results**;
- **Art 13: Transparency and provision of information to deployers:** Clear and adequate information to ensure that their operation is sufficiently transparent to enable **deployers to interpret a system's output and use it appropriately**
- **Art 14: human oversight:** appropriate human-machine **interface tools**, so that they can be **effectively overseen** by natural persons so as **to minimise risk**;
- **Art 15: Accuracy, robustness and cybersecurity**

Conformity assessment of high-risk systems

Step 1



A high-risk AI system is developed.

Step 2



It needs to undergo the conformity assessment and comply with AI requirements.*

*For some systems a notified body is involved too.

Step 3



Registration of stand-alone AI systems in an EU database.

Step 4



A declaration of conformity needs to be signed and the AI system should bear the CE marking.
The system can be placed on the market.

If substantial changes happen in the AI system's lifecycle

Go back to Step 2

Obligations of **developers** of HRS(Sect. 3)

- ensure **compliance with requirements** in Sect. 2
- provide own identification
- have a **quality management** system
- keep the **documentation**
- keep the logs automatically generated
- ensure that the high-risk AI system undergoes the **relevant conformity assessment**
- draw up an EU **declaration of conformity**
- affix the **CE marking**
- comply with the **registration obligations**
- the necessary **corrective actions** and provide **information to authorities**;
- **upon** a reasoned **request of a national competent authority, demonstrate conformity**
- ensure that the high-risk AI system complies with **accessibility requirements**

Risk management system (Art. 9)

Continuous iterative process including

- **identification and analysis risks** to health, safety or fundamental rights arising from intended use
- **estimation and evaluation** of such **risks**;
- evaluation of other risks
- adoption of **appropriate and targeted** risk management **measures**

Risk management measures should ensure:

- (a) **elimination or reduction** of risks as far as technically feasible;
- (b) **mitigation and control** measures for **remaining risks**;
- (c) provision of **information and training** to deployers.

High-risk AI systems should be tested for the purpose of identifying the most appropriate and targeted risk management measures.

Residual risks should be **acceptable**

Obligations of **deployers** of HRS (Sect. 3)

- ensure that systems are **used in accordance with instructions**
- **assign human oversight** to competent individuals
- ensure that **input data is relevant and sufficiently representative**
- **monitor** systems operation, **inform** providers and authorities of **risks or incidents**
- keep logs
- **inform workers' representatives and affected workers** of use of **high-risk AI** system (for employers)
- where applicable, carry out a **data protection impact assessment**, using the information provided by developer
- in the framework of **investigations** for the targeted search of a person involved in criminal offence, **request authorisation by competent judicial or administrative authority**
- if system makes or assist in making decisions related to natural persons, **inform natural persons**
- **cooperate** with the relevant competent **authorities**

Who should certify

- For AI products (or safety components of products) already subject to third party certification (Annex II)
 - Existing certification authorities
- For other AI applications (Annex III)
 - Self certification
 - Certification authorities (notified bodies), for biometric systems not compliant with harmonized standards

Annex II legislation covers:

- Machinery
- Toys
- Lifts
- Radio Equipment
- Personal watercraft
- Medical devices
- Equipment used in potentially explosive atmosphere
- Pressure equipment
- Personally protective equipment
- Cableway installations
- Appliances burning gaseous fuels
- Civil aviation security
- 2- or 3-wheel vehicles and quadricycles
- Agricultural and forestry vehicles
- Marine equipment
- Interoperability of the rail system
- Motor vehicles and their trailers
- Civil aviation

Annex III categories of purposes:

- Biometric identification and categorisation of natural persons
- Management and operations of critical infrastructure
- Education and vocational training
- Employment, workers management and access to self-employment
- Access to and enjoyment of essential private services and public services and benefits
- Law enforcement
- Migration, asylum and border control management
- Administration of justice and democratic processes

What should be certified (Art. 43)?

Compliance with requirement in section 2

- Risk management system (to identify, assess and manage risks)
- Data and Data Governance (test and validate datasets)
- Technical Documentation (to demonstrate compliance)
- Record-Keeping (automated logs)
- Transparency and Provision of Information to Deployers (instruction for use)
- Human Oversight (so that humans can oversee)
- Accuracy, Robustness and Cybersecurity (to ensure consistent performance)

The residual risks should be he **acceptable**

FUNDAMENTAL RIGHT IMPACT ASSESSMENT 1 (Art. 29)

Deployers shall perform an assessment of the impact on **fundamental rights** that the use of such system may produce. They should provide:

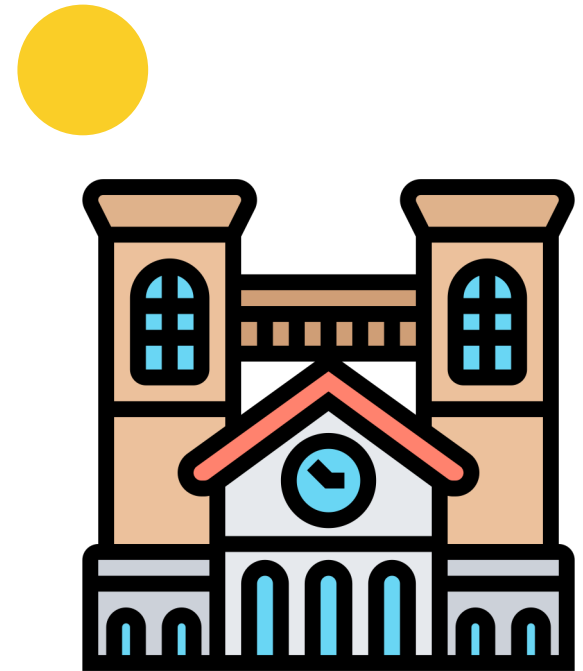
- (a) a **description of the processes** in which the high-risk AI **system will be used**;
- (b) a description of the **period of time of use**;
- (c) the **categories of natural persons and groups** likely to be;
- (d) the **specific risks of harm**
- (e) a **description of the implementation of human oversight measures**;
- (f) the **measures** to be taken in the case of the **materialisation of risks**



EXAMPLES OF WHO NEEDS TO CARRY OUT A FRIA:

- A public employment agency using an AI system to screen job applicants
- A hospital deploying AI for diagnostic support in patient care
- A university using AI to assess student exam performance or allocate admissions
- A housing authority using AI to prioritise applicants for public housing

These assessments must be completed **before putting the AI system into use**, and must examine **how** the system may affect rights such as privacy, equality, non-discrimination, and access to services.



FRAIA



Impact Assessment Fundamental rights and algorithms for governmental organisations

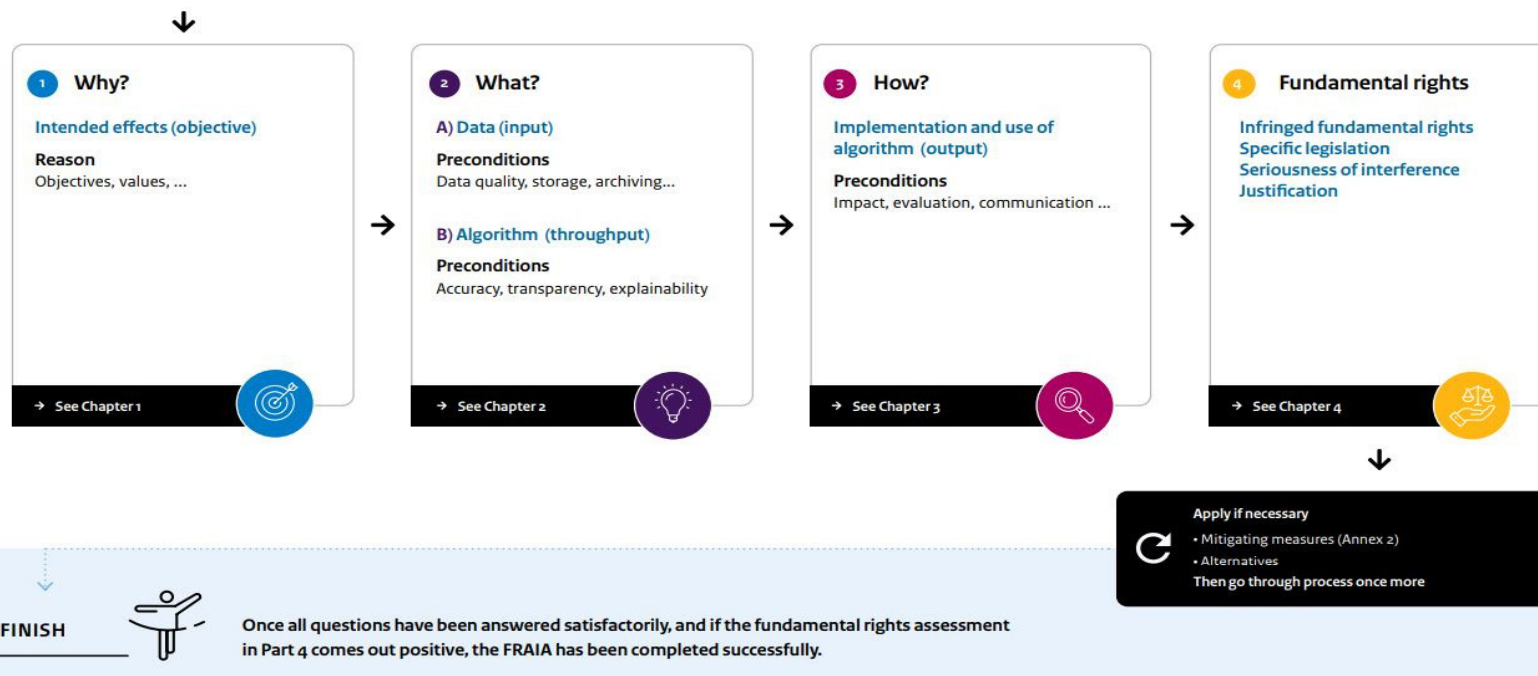
Role	FRAIA Part 1	FRAIA Part 2	FRAIA Part 3	FRAIA Part 4
Interest group	•			
Management	•			
Citizen panel	•			
CISO or CIO	•	•		
Communications specialist		•	•	
Data scientist		•	•	
Data controller or data source owner		•		
Domain expert (employee who has domain knowledge regarding the algorithm's scope of application)	•	•	•	•
Data protection officer		•		
HR staff member			•	
Legal advisor	•	•	•	•
Algorithm developer		•		
Commissioning client	•	•	•	
Other project team members	•			
Project leader	•	•	•	•
Strategic ethics consultant		•	•	

- The FRAIA comprises **a large number of questions** about the topics that need to be discussed and to which an answer must be formulated .
- The discussion about the various questions should take place in a **multidisciplinary team** consisting of people with a wide range of specialisations and backgrounds.
- **Per question, the FRAIA indicates who should be involved in the discussion.**
- However, the **list is not exhaustive**. Likewise, the **role or function names may differ** from one organisation to another.

FRAIA

FRAIA flow chart

START



- <https://www.government.nl/documents/reports/2021/07/31/impact-assessment-fundamental-rights-and-algorithms>

FRAIA



Impact Assessment Fundamental rights and algorithms for governmental organisations

- **Stage 1: preparation.** Why an algorithm will be **used** and what its **effects** will be.
- **Stage 2: input and throughput.** This stage is about the **development** of an algorithmic system. In this stage, it is decided **what the algorithm must look like, and which data is being used to feed** the algorithm. Within this stage, the FRAIA further distinguishes between:
 - **Stage 2a: data, or input.** This involves asking questions that pivot on the use of specific data and data sources;
 - **Stage 2b: algorithm, or throughput.** This involves questions regarding the algorithm, and its operation and transparency.
- **Stage 3: output, implementation and supervision.** This stage is about **how to use the algorithm**, i.e., about the question which output the algorithm generates, how that may play a role in policy or decision-making, and how that can be supervised.

Back to GenAI (I)

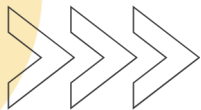
'general-purpose AI model' means an AI model,

- Including where such an AI model is trained with a large amount of data using self-supervision at scale, that
- that displays significant generality and is capable of competently performing a wide range of distinct tasks [...]and
- that can be integrated into a variety of downstream systems or applications [...]; [Art 3 (63)]



'general-purpose AI system' means an AI system

- which is based on a general-purpose AI model and
- which has the capability to serve a variety of purposes, both for direct use as well as for integration in other AI systems;

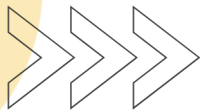


Back to GenAI (II)

‘systemic risk’ means a risk that is specific to the high-impact capabilities of general-purpose AI models, having a significant impact on the Union market

- due to their reach, or
- due to actual or reasonably foreseeable negative effects on public health, safety, public security, fundamental rights, or the society as a whole, that can be propagated at scale across the value chain;

- Article 55: Obligations for Providers of General-Purpose AI Models with Systemic Risk

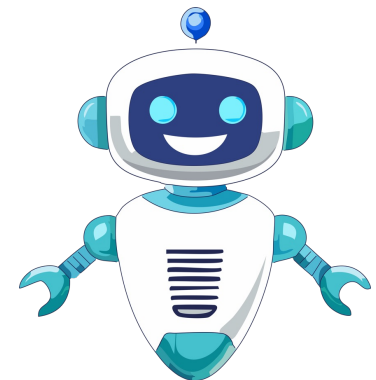


Obligations for providers of GenAI (Art. 53)

- **Technical documentation**
- **Information for providers** of downstream systems
- **Policy** to comply with copyright
- **(Exhaustive) Summary of content** used in training that is specific to the high-impact capabilities of general-purpose AI models
- **Cooperation** with Commission and national authorities
- **Appoint authorised representative**



PS: Commission can adopt delegated acts

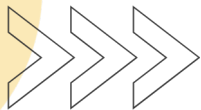
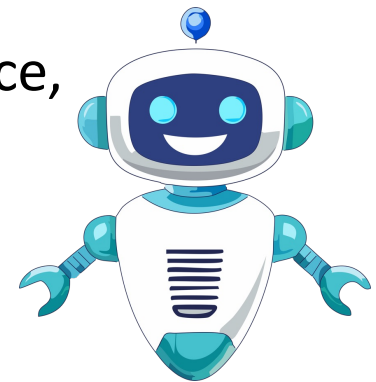


Additional obligations for providers of GenAI with systemic risk (Art. 55)

- Perform **model evaluation**
- **Assess and mitigate systemic risks**
- Keep **track** or, **document and report incidents**
- Ensure **cybersecurity**



Codes of practice (Article 56) may be used to demonstrate compliance, **until harmonized standards are provided**



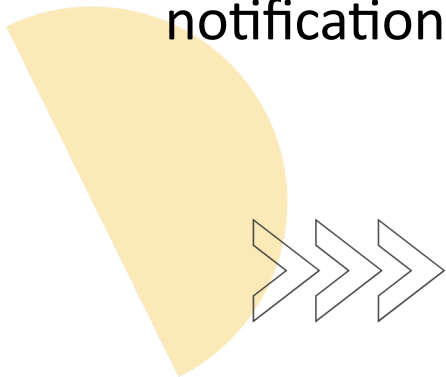
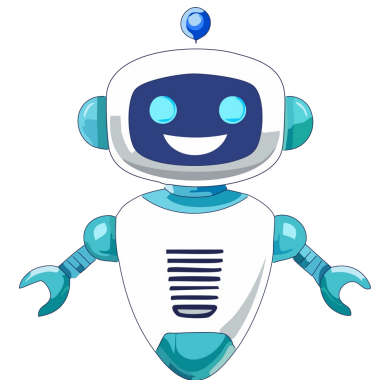
Obligations for providers of GenAI (53)

Obligation	Normal GPAI	GPAI with Systemic Risk
Technical documentation (Annex XI)	✓ Required	✓ Required
Info for downstream AI system providers	✓ Required (Annex XII)	✓ Required (Annex XII)
Public summary of training data	✓ Required	✓ Required
Copyright compliance	✓ Required	✓ Required
Cooperation with regulators	✓ Required	✓ Required
Codes of practice / harmonised standards	✓ Permitted for compliance	✓ Permitted for compliance
Additional risk mitigation (e.g. red teaming, incident reporting, cybersecurity)	✗ Not required	✓ Mandatory
Independent evaluation & model registration	✗ Not required	✓ Required



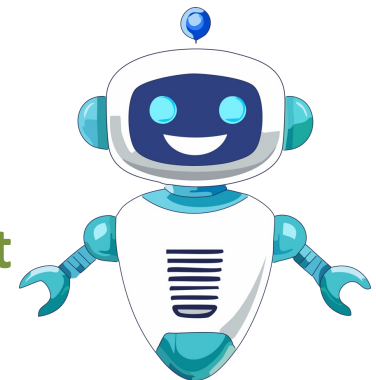
Third-party certification actors

- **Notifying authorities:**
 - **Each Member State** designates or establishes at least one notifying authority
 - The authority is **responsible for setting up and carrying out procedures for the assessment, designation and notification** of conformity assessment bodies and for their monitoring.
- **Certification bodies (notified bodies):**
 - Conformity assessment bodies shall submit an application for notification



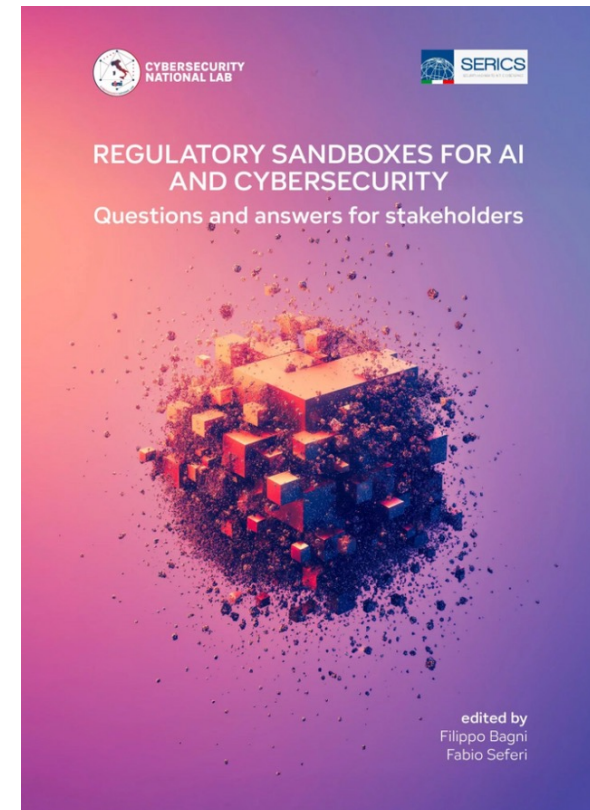
Public actors: EU level

- **Commission:** issues guidelines, delegated acts, implementing acts, **requests/approves harmonised standards**
- **AI Office:** develops Union expertise and capabilities in the field of AI and **contributes to the implementation of Union law on AI.**
- **AI Board:** composed of 1 representative per Member State **advises and assists** the Commission and the Member States in order to facilitate the consistent and effective application of AIA
- **Advisory Forum:** balanced selection of **stakeholders**, including industry, start-ups, SMEs, civil society and academia, **providing technical expertise and advising** the Board and the Commission
- **Scientific Panel:** independent experts **supporting the enforcement activities** under this Regulation.



AI REGULATORY SANDBOXES (Art 57)

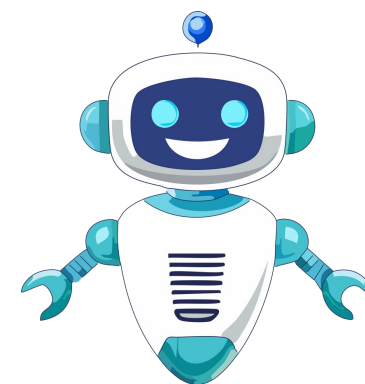
- Voluntary frameworks to **test** AI systems in a **controlled environment** (at least **1 sandbox per member state**)
- Organised by national competent authorities in each Member State
- Open to SMEs, start-ups, and other providers developing innovative AI
- **Must ensure fundamental rights are safeguarded during testing**
- Aim to **support compliance and reduce barriers to innovation**
- Must be operational starting August 2026



Public actors: National level

National competent authorities.

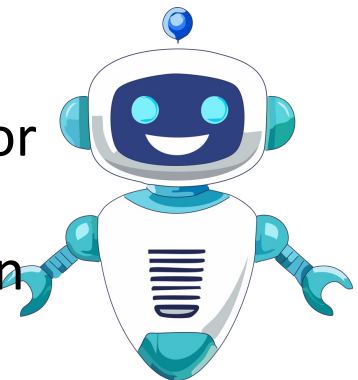
- Each Member State shall establish or designate
 - at least **1 notifying authority** and
 - at least **1 market surveillance authority**
- Member States shall
 - designate a market surveillance authority to act as the **single point of contact**,
 - notify the Commission of the identity of the single point of contact.



Market surveillance authorities in Italy

DDL Intelligenza artificiale

- **l'Agenzia per l'Italia digitale (AgID)** e **l'Agenzia per la cybersicurezza nazionale (ACN)** are designated as the National Authorities for Artificial Intelligence:
 - a) **l'AgID** is responsible for promoting innovation and development of artificial intelligence, without prejudice to the provisions of letter b). AgID is the **notifying authority**;
 - b) **l' Agenzia per la Cybersicurezza Nazionale ACN** is responsible for the supervision, including inspections and sanctions, of artificial intelligence systems. The ACN is also responsible for the promotion and development of artificial intelligence with respect to cybersecurity profiles.



Who punishes whom?

Member state lay down penalties and enforcement measures (Art 99)

- Higher penalties for prohibited practices (Art. 5)
- Penalties for violation of other provisions
- Penalties for providing incorrect, incomplete or misleading information to certification bodies and authorities

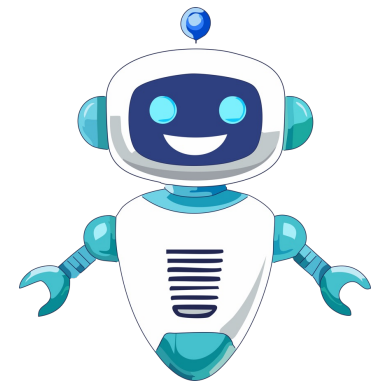


The Commission may impose penalties on providers of general purpose AI models for

- Violations of obligations Art. 53/54
- Failure to provide information or access
- Failure to implement measures



Court of justice has competence to review



ENFORCEMENT



7% or €35M

Prohibited AI violations:
up to 7% of global annual
turnover or 35 million
euros



3% or €15M

**Most other violations
(providers and
deployers):** up to 3% of
global annual turnover or
15 million euros.



1% or €7.5M

**Supplying incorrect
information to
authorities:** up to 1% of
global annual turnover or
7.5 million euros.

- **National supervisory authorities** enforce the AI Act in each Member State
- **The European Artificial Intelligence Office** supports coordination and oversight
- Fines can reach up to €35 million or 7% of global annual turnover

SUMMARY (AI ACT)

- The first comprehensive legal framework for regulating AI globally
- Focuses on AI safety and the protection of fundamental rights
- Applies a risk-based approach, with stricter rules for high-risk AI systems
- Covers the entire AI lifecycle, from development to deployment and use



Sources:



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